

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0197

Cub-Run Dp 138 kV
September 23, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0197. The Interconnection Customer proposes a 183.7 MW wind facility interconnecting at the Cub-Run Dp 138 kV substation in the Virginia area. The request was received on January 16, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 131 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,052,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Colonial – Idaho Springs 138 KV Line
Contingency: Colonial – Sideburn 138 KV
Loading: 141.7% (Rating: 302 MVA)
DFAX: 9.3%
Pre-project loading: 136.1%
2. Facility: Colonial – Sideburn 138 KV
Contingency: Colonial – Idaho Springs 138 KV
Loading: 136.5% (Rating: 265 MVA)
DFAX: 6.7%
Pre-project loading: 131.8%
3. Facility: Cub-Run Dp – Grand Blanc Generating Station Substation 138 KV line
Contingency: Idaho Springs Substation – Colonial 138 KV
Loading: 125.8% (Rating: 239 MVA)
DFAX: 52.9%
Pre-project loading: 85.2%
4. Facility: Norfolk – Battery Hts Dp 138 KV Circuit 1
Contingency: Batter Hts Dp – Norfolk 138 KV #2
Loading: 123.4% (Rating: 333 MVA)
DFAX: 10.2%
Pre-project loading: 117.7%
5. Facility: Clifton – Norfolk 138 KV #2
Contingency: Norfolk Substation – Clifton 138 KV Circuit 1
Loading: 121.7% (Rating: 401 MVA)
DFAX: 9.2%
Pre-project loading: 117.5%

6. Facility: Norfolk Substation – Battery Hts Dp 138 KV Ckt 2
Contingency: Norfolk – Battery Heights Dp 138 KV Circuit 1
Loading: 121.1% (Rating: 339 MVA)
DFAX: 10.2%
Pre-project loading: 115.5%
7. Facility: Clifton – Norfrk Substation 138 KV
Contingency: Clifton – Norfolk 138 KV #2
Loading: 118.6% (Rating: 412 MVA)
DFAX: 9.2%
Pre-project loading: 114.5%
8. Facility: Battery Hts Dp – Toa Vaca 138 KV Circuit 2
Contingency: Toa Vaca – Battey Hts Dp 138 KV Circuit 1
Loading: 116.3% (Rating: 334 MVA)
DFAX: 9.6%
Pre-project loading: 111.0%
9. Facility: Toa Vaca Substation – Battery Hts Dp 138 KV Cir. 1
Contingency: Battery Hts Dp – Toa Vaca 138 KV Ckt 2
Loading: 115.0% (Rating: 338 MVA)
DFAX: 9.6%
Pre-project loading: 109.7%
10. Facility: Colonial – Clifton 138 KV Circuit 1
Contingency: Clifton – Coloial 138 KV
Loading: 111.3% (Rating: 419 MVA)
DFAX: 9.0%
Pre-project loading: 107.4%
11. Facility: Colonial – Clifon 138 KV Cir. 2
Contingency: Colonial – Clifton 138 KV
Loading: 111.3% (Rating: 419 MVA)
DFAX: 9.0%
Pre-project loading: 107.4%
12. Facility: Grand Blanc Generating Sta – Del Amo 138 KV
Contingency: Dulles – Lunenburg 138 KV
Loading: 108.4% (Rating: 241 MVA)
DFAX: 13.7%
Pre-project loading: 98.0%
13. Facility: Colesville – Sunbright 138 KV
Contingency: Adrian Energy Associates Llc 345-138 kv XFMR
Loading: 102.6% (Rating: 322 MVA)
DFAX: 6.5%
Pre-project loading: 98.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Colonial – Idaho Springs 138 KV Line	Rebuild line with higher-capacity conductor	\$1,034,000	33% (\$343,000)
Sidebrn – Colonial 138 KV L	Rebuild line with higher-capacity conductor	\$7,318,000	48% (\$3,505,000)
Cub-Run Dp – Grand Blanc Generating Station Substation 138 KV line	Rebuild line with higher-capacity conductor	\$3,870,000	89% (\$3,445,000)
Norfork – Battery Hts Dp 138 KV line	Rebuild line with higher-capacity conductor	\$1,356,000	66% (\$898,000)
Clifton – Norfork 138 KV #2	Rebuild line with higher-capacity conductor	\$2,240,000	40% (\$894,000)
Norfork Substation – Battery Hts Dp 138 KV Ckt 2	Rebuild line with higher-capacity conductor	\$554,000	66% (\$367,000)
Norfork – Clifton 138 KV Circuit 1	Rebuild line with higher-capacity conductor	\$2,240,000	40% (\$895,000)
Battery Hts Dp – Toa Vaca 138 KV Line 2	Rebuild line with higher-capacity conductor	\$1,452,000	66% (\$962,000)
Toa Vaca Substation – Battery Hts Dp 138 KV Cir. 1	Rebuild line with higher-capacity conductor	\$1,030,000	66% (\$682,000)
Colonial – Clifton 138 KV Circuit 1	Rebuild line with higher-capacity conductor	\$1,904,000	40% (\$760,000)
Clifton – Colonial 138 KV #2	Rebuild line with higher-capacity conductor	\$1,911,000	40% (\$763,000)
Grand Blanc Generating Sta – Del Amo 138 KV	Reconductor line	\$517,000	100% (\$517,000)
Colesville – Sunbright 138 KV	Reconductor line	\$433,000	5% (\$21,000)
	Total allocated to project		\$14,052,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.